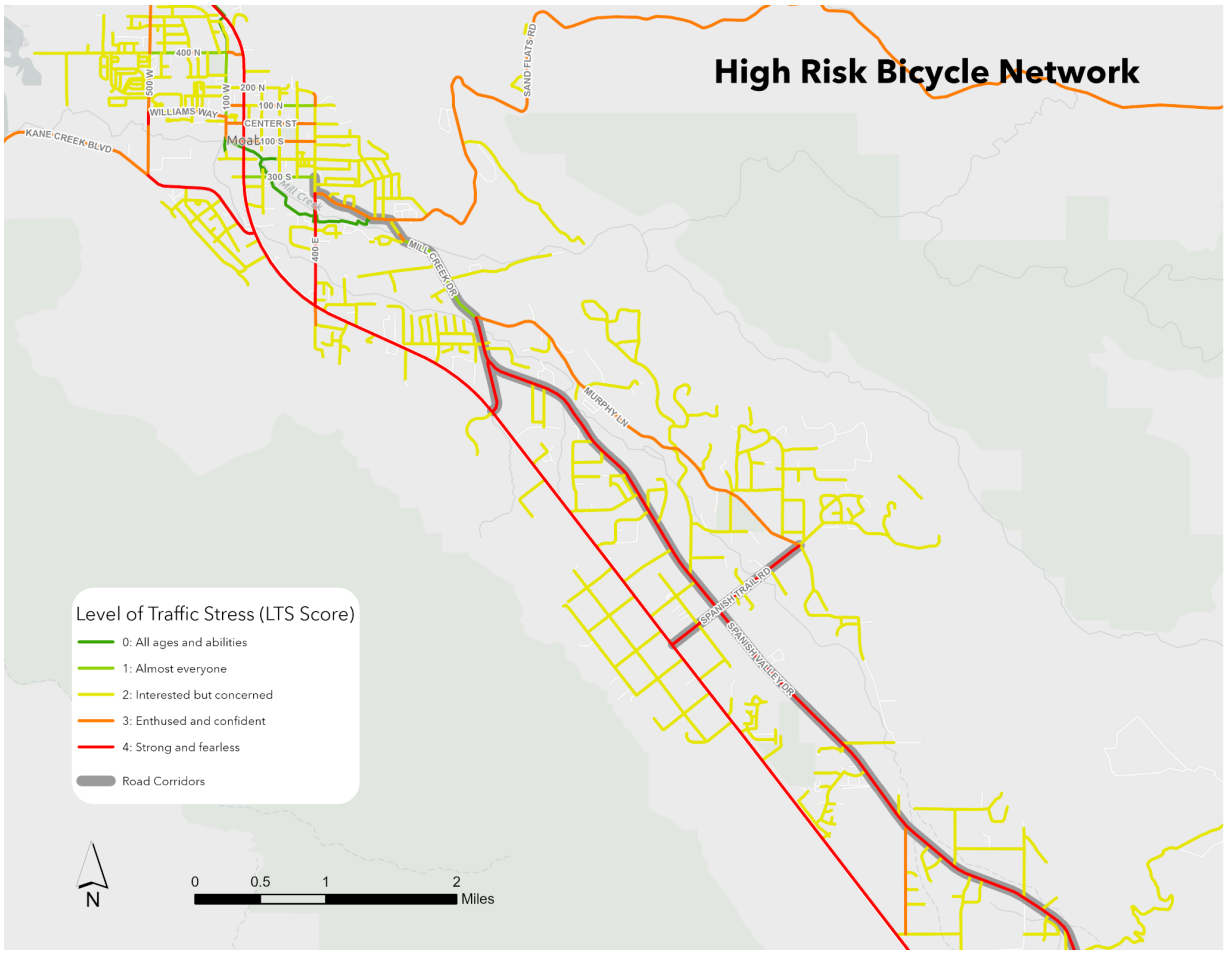
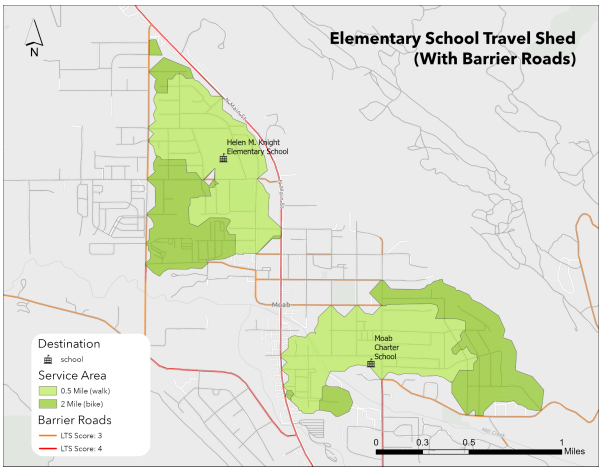
Carson Miller | (734) 756-0392 | carsonmillerw@gmail.com

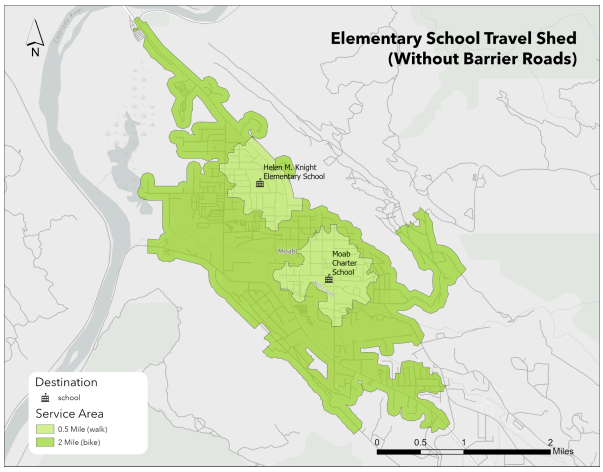
GIS Product Example: Bike Utah
Moab Safe Streets For All Plan: Travel Shed Analysis and Equity Index Mapping

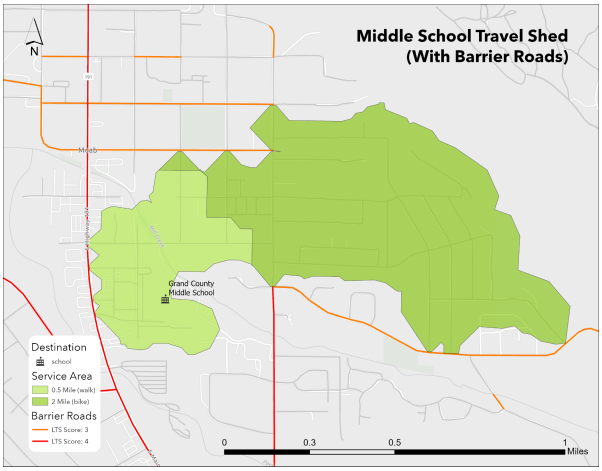
Travel Shed Analysis

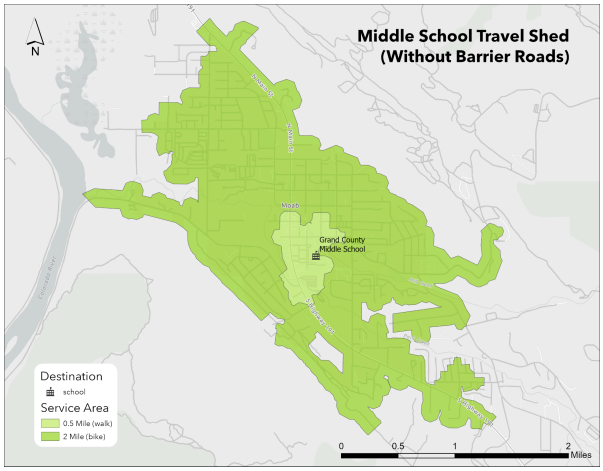


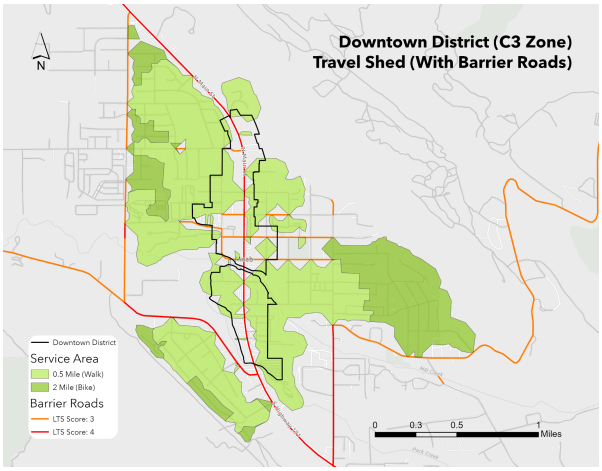
Using high-stress roadways (LTS 3 and 4) as network barriers, I conducted travel shed analyses comparing service areas for a 0.5 mile walking radius and a 2 mile cycling radius. I then quantified the resident populations within each service area to measure the impact these barrier roads have on accessibility.

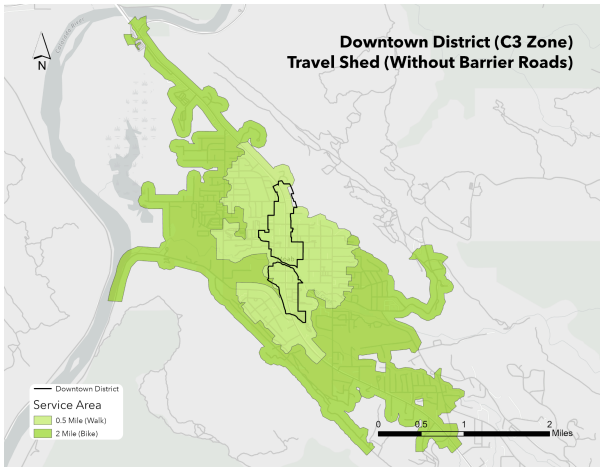

Elementary School Travel Shed
(With Barrier Roads)
N
Helen M. Knight Elementary School
Moab Charter School
Moab
Destination
school
Service Area
0.5 Mile (walk)
2 Mile (bike)
Barrier Roads
LTS Score: 3
LTS Score: 4
0
0.3
0.5
1
Miles

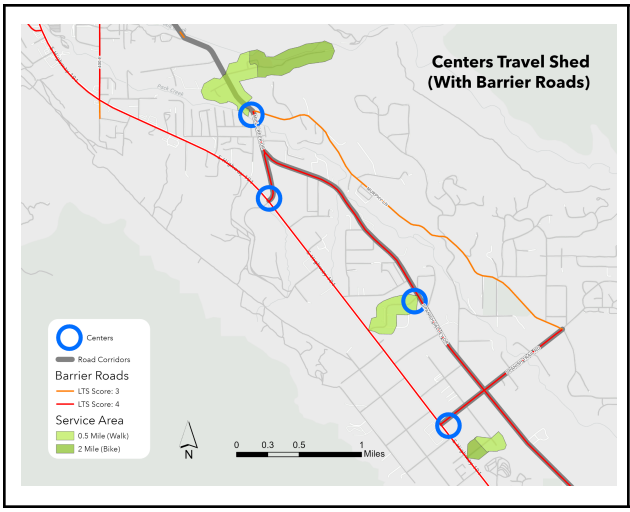

Elementary School Travel Shed
(Without Barrier Roads)
N
Helen M. Knight Elementary School
Moab Charter School
Moab
Destination
school
Service Area
0.5 Mile (walk)
2 Mile (bike)
0
0.5
1
2
Miles

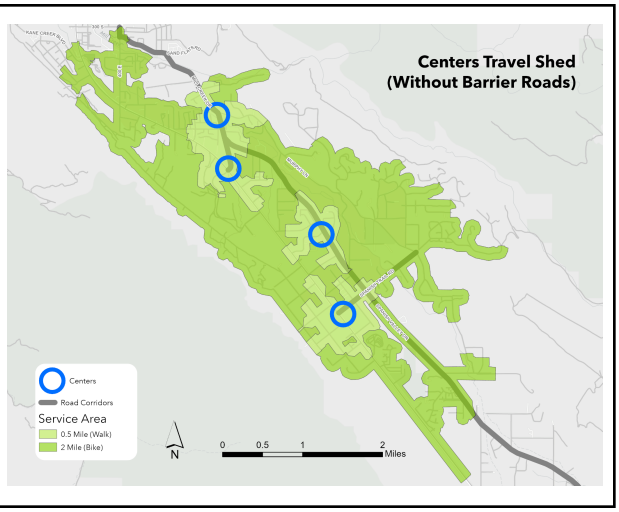

Middle School Travel Shed
(With Barrier Roads)
N
Grand County Middle School
Moab
Destination
school
Service Area
0.5 Mile (walk)
2 Mile (bike)
Barrier Roads
LTS Score: 3
LTS Score: 4
0
0.3
0.5
1
Miles

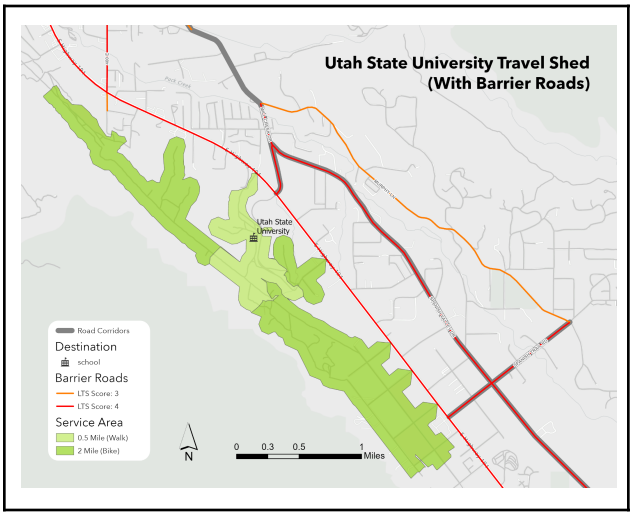

Middle School Travel Shed
(Without Barrier Roads)
N
Grand County Middle School
Moab
Destination
school
Service Area
0.5 Mile (walk)
2 Mile (bike)
0
0.5
1
2
Miles

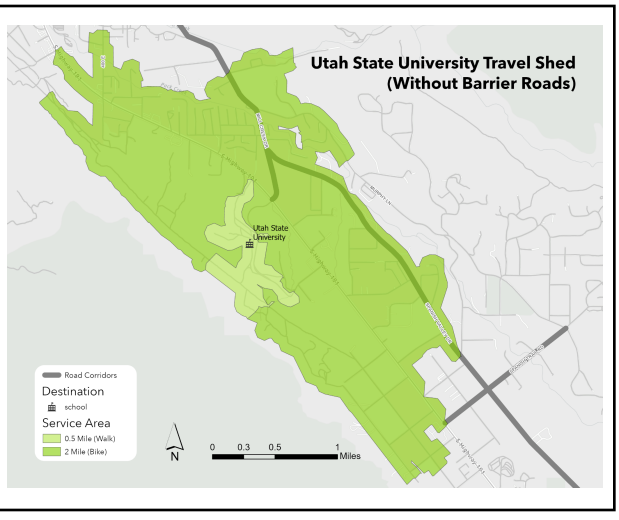

Downtown District (C3 Zone)
Travel Shed (With Barrier Roads)
N
Downtown District
Service Area
0.5 Mile (Walk)
2 Mile (Bike)
Barrier Roads
LTS Score: 3
LTS Score: 4
0
0.3
0.5
1
Miles


Downtown District (C3 Zone)
Travel Shed (Without Barrier Roads)
N
Downtown District
Service Area
0.5 Mile (Walk)
2 Mile (Bike)
0
0.5
1
2
Miles





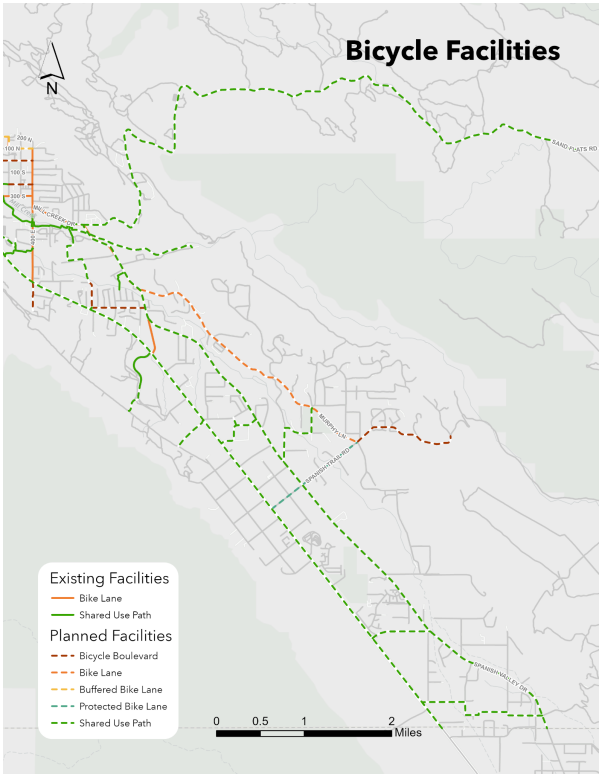


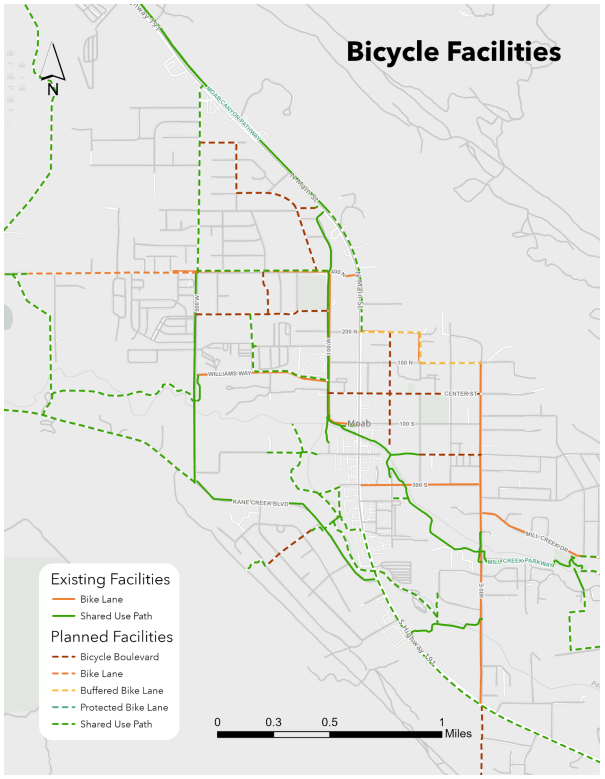


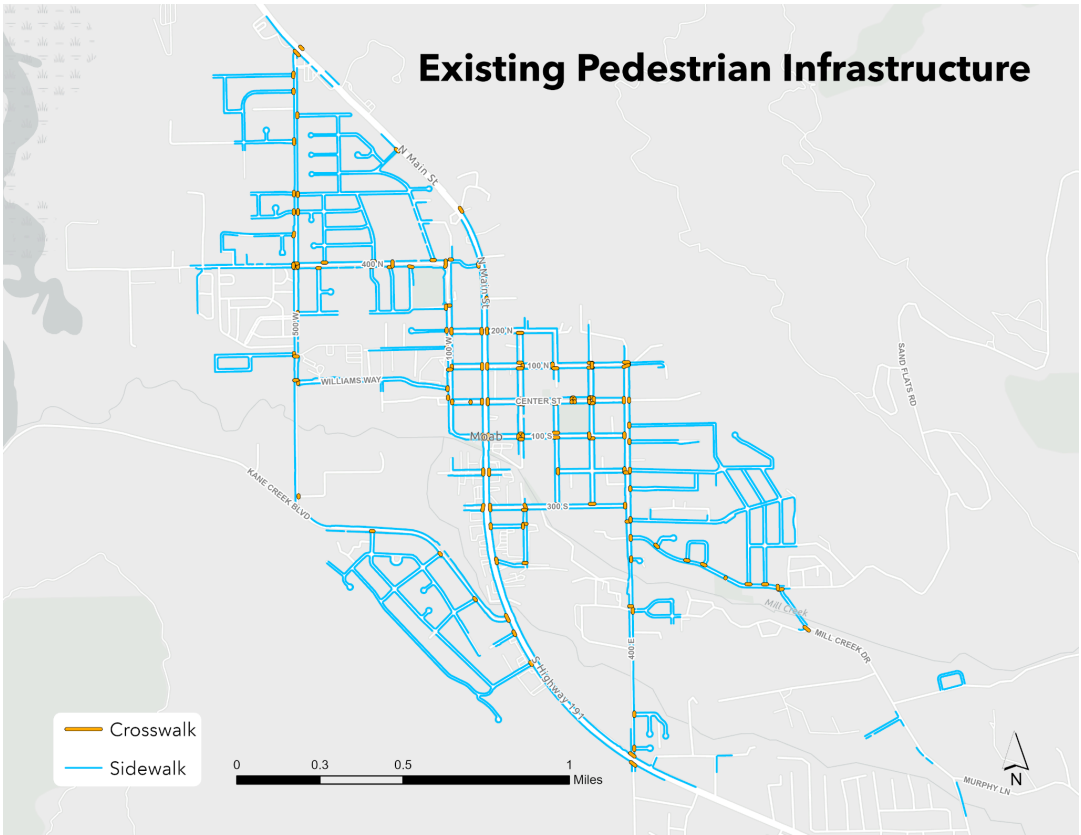
Destination/Cut Off	With Barrier	Without Barrier	Change	% Change
Elementary Schools				
0.5 Mile	758	1661	903	119.13%
2 Mile	1146	3978	2832	247.12%
Middle School				
0.5 Mile	145	217	72	49.66%
2 Mile	740	4854	4114	555.95%
Downtown District (C3 Zone)				
0.5 Mile	1702	2499	797	46.83%
2 Mile	837	3313	2476	295.81%
Centers				

0.5 Mile	99	992	893	902.02%
2 Mile	24	2171	2147	8945.83%
Utah State University				
0.5 Mile	2	2	0	0%
2 Mile	188	1820	1632	868.09%

Existing and Planned AT Facilities







Underrepresented and Marginalized Populations				
Category	**Definition**	**Source Table (2024 ACS 5-Year Estimates)**	**Grand County**	**Moab City**
Age (youth)	Residents under 18 years old	B09001	1,884 19.3%	1,172 22.1%
Age (elderly)	Residents 65 years old or older	B01001	1,958 20.1%	941 17.7%
Income	Residents* in households with income under $49,999 (75% AMI)	S1901 B25010	3,507 36%	2094 39%
Race/Ethnicity	Residents who are non-white and/or Hispanic	B03002	2,181 22.4%	1,533 28.9%
English Proficiency	Residents with limited English proficiency (speaks English less than “very well”)	C16001	346 3.5%	302 5.7%
Disability	Residents with a disability	B18101	1,373 14.1%	659 12.4%
Vehicle Access	Residents* in households with zero vehicles	B25044 B25010	532 5.5% (252 units)	456 8.6% (214 units)
Housing availability	Residents* living in crowded households (HUD definition of crowding: Occupied units with more than 1 occupant per room)	B25014 B25010	411 4.2% (195 units)	343 6.5% (161 units)

*Calculated by multiplying the number of households and the average household size

- Grand County (B25010): Average household size: 2.11
- Moab City (B25010): Average household size: 2.13

EQUITY INDEX MAPPING

This analysis identifies areas of higher need in Grand County and Moab by mapping historically marginalized or underrepresented population groups. The purpose of this Equity Priority Index is to highlight where transportation investments can most effectively improve safety and access for those with the greatest need.

Methodology

This Equity Priority Index is based on the methodology developed by the Wasatch Front Regional Council (WFRC) for its 2024 Comprehensive Safety Action Plan, adapted to reflect conditions in Moab. It combines multiple demographic and socioeconomic indicators into a single measure to identify priority areas for investment and planning.

$$Index = \frac{(Yth + Eld + NH + LEP + Dis + (Pov * HH) + (Veh * HH) + (Crwd * HH)}{Pop}$$

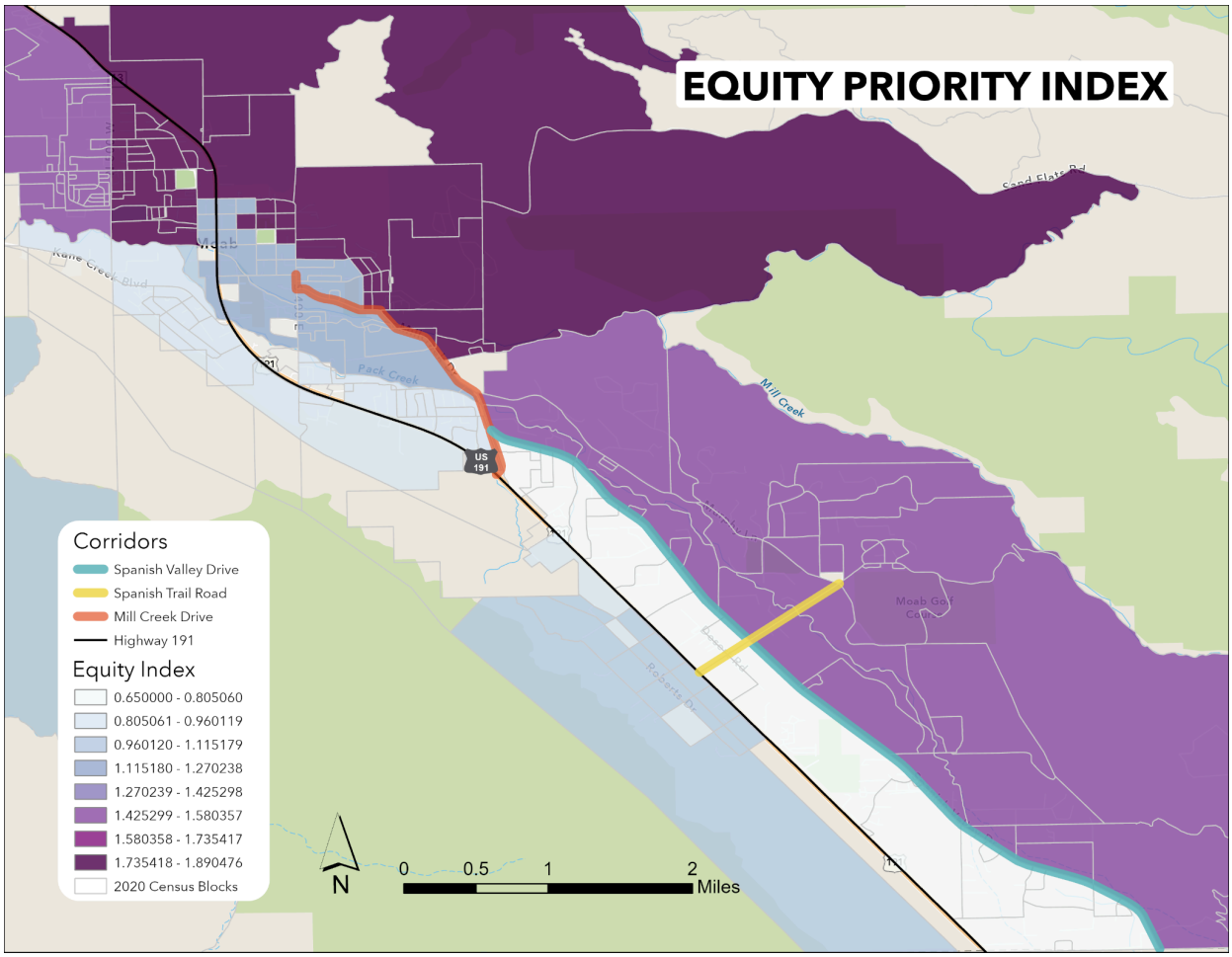
Where the variables represent:

Yth: Number of residents under 18 years old
Eld: Number of residents over 65 years old
NH: Number of non-white and/or hispanic residents
LEP: Number of residents with limited English proficiency
Dis: Number of residents with a disability
Pov: Number of households with income below $49,999
Veh: Number of households without zero vehicles
Crwd: Number of households with more than 1 occupant per room
HH: Average household size in Moab

This analysis uses 2020 Census blocks as the base geography to allow comparisons across neighborhoods. While block-level Census data provide population, age, and race/ethnicity, additional variables required for the equity priority index, such as income, vehicle access, and disability, are drawn from the 2024 American Community Survey (ACS) 5-year estimates, which are only available at the larger block group level.

To estimate these values at the block level, block group data were distributed to individual blocks based on population. In other words, blocks were assigned a share of each demographic characteristic proportional to their share of the block group population, using 2020 Census data. This approach assumes that these characteristics are spread evenly within each block group and that population patterns have not changed significantly between 2020 and 2024. As a result, block-level values are estimates intended to show general patterns rather than exact counts.

Below shows the resulting GIS products from the Equity Index evaluation.


EQUITY PRIORITY INDEX
Corridors
Spanish Valley Drive
Spanish Trail Road
Mill Creek Drive
Highway 191
Equity Index
0.650000 - 0.805060
0.805061 - 0.960119
0.960120 - 1.115179
1.115180 - 1.270238
1.270239 - 1.425298
1.425299 - 1.580357
1.580358 - 1.735417
1.735418 - 1.890476
2020 Census Blocks
N
0
0.5
1
2
Miles
Pack Creek
Mill Creek
US 191